

Examples of IPMs

	Function Class $\mathcal{F}$	IPM Form
1-Wasserstein	$\{f : \text{Lip}(f) \leq 1\}$	$\sup_{f: \text{Lip}(f) \leq 1} \left(\mathbb{E}_\alpha[f] - \mathbb{E}_\beta[f] \right)$
Total Variation (TV)	$\{f : \ f\ _\infty \leq 1\}$	$\sup_{\ f\ _\infty \leq 1} \left(\mathbb{E}_\alpha[f] - \mathbb{E}_\beta[f] \right) = \frac{1}{2} \int d\alpha - d\beta $
MMD	$\{f : \ f\ _{\mathcal{H}_k} \leq 1\}$	$\sup_{\ f\ _{\mathcal{H}_k} \leq 1} \left(\mathbb{E}_\alpha[f] - \mathbb{E}_\beta[f] \right)$
Energy Distance	MMD with certain kernels (eg $k(x, y) = -\ x - y\ $)	$\sup_{f \in \mathcal{F}_{ED}} \left(\mathbb{E}_\alpha[f] - \mathbb{E}_\beta[f] \right)$
Flat/Dudley Metric	$\mathcal{F}_{\text{flat}} = \{f : \text{Lip}(f) \leq 1, \ f\ _\infty \leq 1\}$	$\sup_{f \in \mathcal{F}_{\text{flat}}} \left(\mathbb{E}_\alpha[f] - \mathbb{E}_\beta[f] \right)$

Maximum Mean Discrepancy